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Diacritic-Aware Post-OCR Correction for Vietnamese Scanned Documents: A Lightweight Baseline for Low-Quality Document Digitization

Nguyễn Tuệ An

2206090055@ms.hanu.edu.vn

Hanoi University https://orcid.org/0009-0001-4914-6647

Trần Thị Lan

Hanoi University

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Diacritic-Aware Post-OCR Correction for Vietnamese Scanned Documents: A Lightweight Baseline for Low-Quality Document Digitization

Nguyễn Tuệ An,[1] Trần Thị Lan[1]

[1] *Faculty of Information Technology (FIT), Hanoi University (HANU), Hanoi, Vietnam*

Abstract:
Optical Character Recognition (OCR) is an important technology for converting scanned documents into machine-readable text. However, Vietnamese OCR remains challenging because Vietnamese uses a Latin-based writing system with complex diacritics and tonal marks. These marks are often small and visually sensitive to blur, low resolution, uneven illumination, compression artifacts, and degraded printing quality. As a result, OCR systems may recognize the base characters correctly while missing, confusing, or misplacing Vietnamese diacritics, which can significantly change the meaning of words. For example, *ban*, *bán*, *bàn*, *bản*, and *bạn* share similar base forms but have different meanings.
This paper presents a lightweight diacritic-aware post-OCR correction pipeline for Vietnamese scanned documents. The proposed method combines simple image preprocessing, OCR recognition, Vietnamese-specific error analysis, dictionary-based correction, and diacritic restoration rules. Unlike heavy end-to-end OCR models that require large annotated datasets and expensive training, the proposed baseline is designed for practical document digitization scenarios where data and computational resources are limited. We evaluate the pipeline on a collection of Vietnamese scanned document images containing different layouts, font sizes, noise levels, and document types. The evaluation uses character error rate, word error rate, diacritic error rate, and exact word accuracy. In addition, we provide qualitative analysis of common Vietnamese OCR errors and sample document-level correction results. The proposed work provides a simple and reproducible baseline for Vietnamese OCR improvement and can support future research in low-quality document digitization, historical document analysis, and low-resource language technology.

1. Introduction

The digitization of printed and scanned documents is increasingly important for education, administration, cultural preservation, digital libraries, and information retrieval. OCR systems make it possible to convert scanned images into editable and searchable text. For high-resource languages such as English, OCR technology has achieved strong performance in many practical settings. However, OCR for Vietnamese still faces important challenges, especially when documents are captured under low-quality conditions.

Vietnamese uses a Latin-based script but contains a rich system of diacritics and tone marks. A small change in a diacritic can produce a completely different word. For example, the forms *ma*, *má*, *mà*, *mã*, *mả*, and *mạ* are visually similar but semantically different. Similarly, *ban*, *bán*, *bàn*, *bản*, and *bạn* may be confused when tone marks are missing or incorrectly recognized. This makes Vietnamese OCR different from OCR for non-tonal Latin-script languages, because recognizing only the base letters is not enough.

In low-quality scanned documents, diacritics are often the first elements to be lost. Small marks above or below vowels may disappear due to blur, compression, weak contrast, or image binarization. OCR engines may also confuse Vietnamese characters such as *ă*, *â*, *ê*, *ô*, *ơ*, *ư*, and their tonal variants. These errors reduce the usefulness of OCR outputs for downstream tasks such as search, text mining, document retrieval, translation, plagiarism detection, and archival preservation.

Recent Vietnamese document analysis studies have shown the effectiveness of combining text detection and OCR recognition models such as Differentiable Binarization and VietOCR for structured document understanding [46]. Other OCR-related work has integrated OCR with multilingual NLP for English–Vietnamese plagiarism detection [47]. However, many practical digitization scenarios still require a lightweight correction stage after OCR, especially when the input images are degraded or when retraining OCR models is not feasible. Inspired by low-resource OCR correction strategies [25, 48], this paper proposes a simple but effective diacritic-aware post-OCR correction baseline for Vietnamese scanned documents.

The main contributions of this paper are:

1. We present a lightweight Vietnamese post-OCR correction pipeline focusing on diacritic and tone-mark errors.
2. We define a Vietnamese OCR error taxonomy covering missing diacritics, wrong tone marks, vowel-shape confusion, character substitution, word segmentation errors, and layout-related noise.
3. We propose a dictionary-based and rule-based correction strategy that can be used without large-scale model training.
4. We provide document sample analysis, tables, and evaluation metrics suitable for reproducible Vietnamese OCR experiments.

2. Related Work

2.1. OCR for Vietnamese Documents

Vietnamese OCR has received increasing attention because of the need to digitize administrative forms, educational records, ID cards, receipts, books, and historical documents. Vietnamese text recognition is more complex than English text recognition because Vietnamese contains multiple diacritic combinations. In addition, Vietnamese documents often include mixed content such as numbers, English terms, abbreviations, names, addresses, and tabular layouts.

Hoang et al. [46] proposed an information extraction system for Vietnamese student ID cards using MobileNetV3-Small for card classification, Differentiable Binarization for text detection, and VietOCR for recognition. Their results showed strong recognition performance in a controlled student-card domain, but real-world scanned documents remain difficult because they may contain skew, blur, uneven lighting, old fonts, low contrast, and compression artifacts.

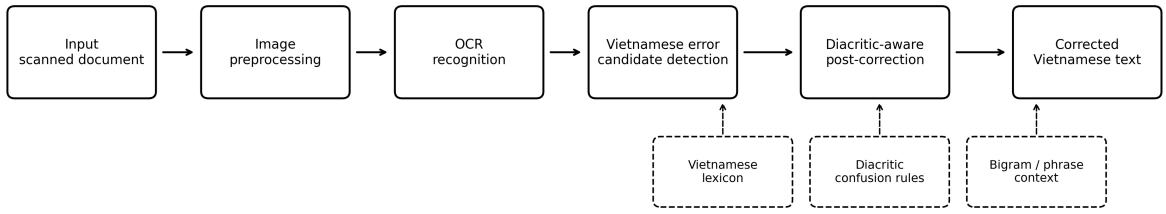


Figure 1: Lightweight diacritic-aware Vietnamese post-OCR correction pipeline.

2.2. Post-OCR Correction

Post-OCR correction is the process of improving OCR output after recognition. Instead of modifying the OCR engine, a correction module analyzes the recognized text and repairs likely errors. Post-OCR correction is useful when retraining the OCR engine is expensive or impossible. Common methods include dictionary lookup, edit distance, character-level language models, n-gram statistics, rule-based substitution, and neural sequence-to-sequence correction.

For low-resource languages, lightweight methods remain valuable because large aligned datasets of noisy OCR and corrected text are often unavailable. Tran et al. [48] demonstrated that a heuristic-based OCR correction pipeline can improve recognition accuracy for a low-resource language setting. This motivates the design of transparent and language-aware post-processing modules for Vietnamese scanned documents.

2.3. Vietnamese Diacritic Ambiguity

Vietnamese diacritics are not only orthographic decorations; they encode lexical information. Removing or changing a tone mark may produce another valid Vietnamese word. Therefore, post-OCR correction for Vietnamese must be careful. A correction system should not blindly add diacritics based only on frequency, because some base forms correspond to many valid words. For example, the base form *ban* may correspond to *ban*, *bán*, *bàn*, *bản*, *bạn*, and other forms. Correct restoration requires context, dictionary constraints, and sometimes document-domain information.

Vietnamese scene-text research further confirms that diacritic ambiguity is a major language-specific challenge. Nguyen et al. [49] reported that Vietnamese vocabulary may contain many diacritic collision groups, where multiple valid words share the same stripped base form. This reinforces the need for diacritic-aware OCR correction. Related work in low-resource manuscript recognition also shows that language-specific units and augmentation strategies can improve recognition robustness when training data are limited [34,45]. In document understanding, fine-grained layout datasets for low-resource scripts further demonstrate that OCR correction should be considered together with layout structure, region grouping, and reading order [36].

2.4. Broader Document Analysis and Low-Resource OCR

Modern document analysis increasingly combines text detection, OCR recognition, layout modeling, and post-processing. Scene-text detection methods such as EAST, CRAFT, DBNet, TextSnake, and PSENet provide strong foundations for locating text regions in natural and document images [12–16]. Recognition models such as CRNN, attention-based OCR, SVTR, and TrOCR have improved sequence recognition for printed and scene text [4,5,7,8]. For document understanding, models and datasets such as FUNSD, PubLayNet, LayoutLM, LayoutLMv2, LayoutLMv3, DocVQA, Donut, and CORD show that layout information is essential for extracting structured information from documents [32, 33, 35, 37–41]. These studies support the motivation of this work: Vietnamese OCR correction should not only restore characters but also consider document quality, layout organization, and downstream usability.

3. Proposed Method

The proposed pipeline consists of five stages: image preprocessing, OCR recognition, error candidate detection, diacritic-aware correction, and final text reconstruction.

3.1. Overall Pipeline

Figure 1 illustrates the proposed Vietnamese post-OCR correction pipeline. The pipeline is modular: the OCR engine can be Tesseract, VietOCR, PaddleOCR, EasyOCR, or any Vietnamese-compatible OCR system. The correction module is independent and can be applied to the output of any OCR engine.

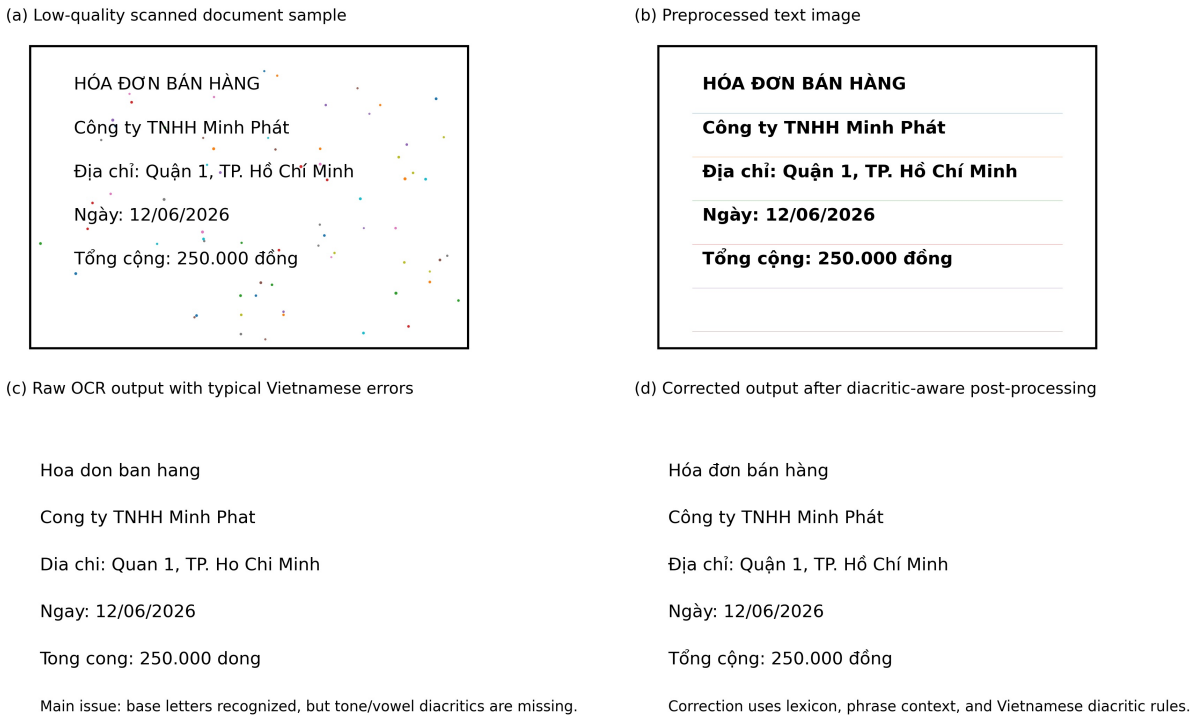


Figure 2: Sample document analysis for Vietnamese post-OCR correction.

3.2. Image Preprocessing

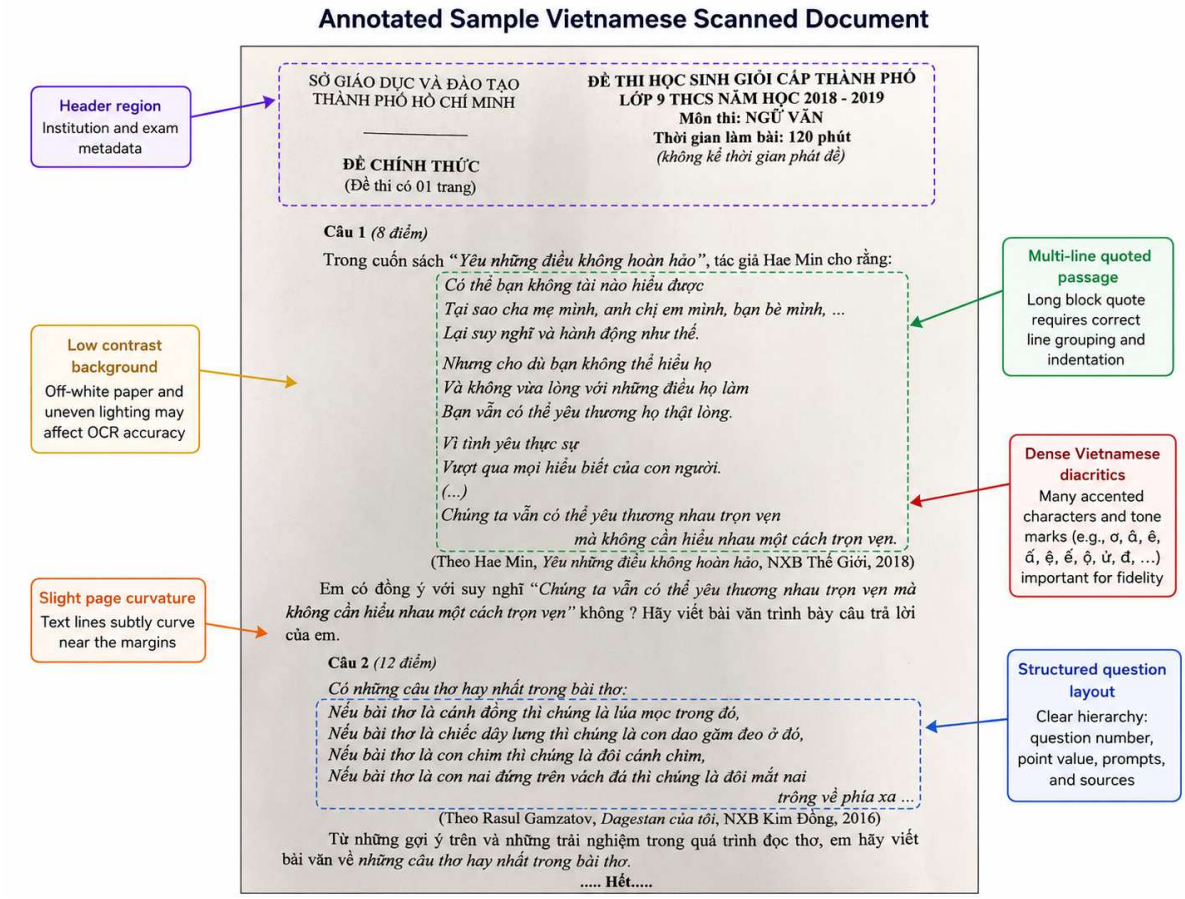


Figure 3: Annotated Vietnamese exam-page sample showing OCR difficulty factors and document-structure regions.

Before OCR recognition, the input image is processed to improve text visibility. The preprocessing stage may include grayscale conversion, denoising, contrast enhancement, adaptive thresholding or Otsu binarization, deskewing, border removal, and table or text-line region cropping when necessary.

For low-quality scanned documents, preprocessing is important because Vietnamese diacritics are small and easily degraded. Over-aggressive binarization may remove tone marks, while insufficient preprocessing may leave background noise that causes false recognition.

3.3. OCR Recognition

After preprocessing, the document image is passed to an OCR engine. In this study, the OCR engine is treated as a black-box recognizer. The raw OCR output is preserved for comparison with the corrected output.

Let the OCR output be represented as:

$$T_{\text{raw}} = \{w_1, w_2, \ldots, w_n\}, \tag{1}$$

where each w_i is a recognized token. The goal of post-OCR correction is to produce:

$$T_{\text{corr}} = \{\hat{w}_1, \hat{w}_2, \ldots, \hat{w}_n\}, \tag{2}$$

where $\hat{w}_i$ is the corrected token.

3.4. Vietnamese Error Candidate Detection

Not every OCR token should be corrected. Some recognized tokens may already be correct, while others may be names, abbreviations, numbers, or English terms. To avoid over-correction, we first identify suspicious tokens.

A token is considered a correction candidate if it satisfies one or more of the following conditions:

1. It is not found in the Vietnamese dictionary.
2. Its base form without diacritics matches multiple Vietnamese dictionary words.
3. It contains unusual character combinations.
4. It has missing tone marks in a context where a diacritic form is more likely.
5. It appears near other OCR errors in the same line.
6. It contains visually similar character confusion, such as *o/ô/ơ*, *a/ă/â*, *u/ư*, or *e/ê*.

3.5. Diacritic-Aware Correction

The correction module uses a candidate-generation and candidate-ranking strategy. First, for each suspicious token, we remove diacritics to obtain its base form:

$$b = \text{strip}(w_i). \tag{3}$$

Then, all valid Vietnamese dictionary words that share the same base form are retrieved:

$$C(w_i) = \{v \in D : \text{strip}(v) = b\}, \tag{4}$$

where D is the Vietnamese dictionary.

If the candidate set is empty, the system applies edit-distance search to find visually similar words. If the candidate set contains multiple words, the system ranks them using character similarity, diacritic similarity, word frequency, local context score, domain dictionary score, and bigram or trigram probability. The final correction is selected as:

$$\hat{w}_i = \arg\max_{c \in C(w_i)} \text{Score}(c, \text{context}). \tag{5}$$

3.6. Correction Rules

Table 1 shows common Vietnamese OCR error types and correction strategies.

Table 1: Vietnamese OCR error taxonomy and correction strategy.

Error Type	Raw OCR Example	Correct Form	Cause	Correction Strategy
Missing tone mark	hoa don	hóa đơn	Tone mark removed by blur	Base-form lookup + context
Wrong tone mark	hỏa đơn	hóa đơn	Similar visual accent	Candidate ranking
Vowel-shape confusion	truong hoc	trường học	Missing ư/ơ	Vietnamese vowel restoration
Broken diacritic	cong nghe	công nghệ	Low resolution	Dictionary and n-gram check
Character substitution	l / 1, O / 0	lớp / lóp	Similar glyphs	Rule-based replacement
Word segmentation error	thanhpho	thành phố	OCR merges words	Dictionary splitting
Layout noise	page-number, header	removed	Header/footer recognized as text	Layout filtering
Table mis-ordering	columns mixed	correct order	OCR ignores table structure	Region-based OCR

4. Dataset and Experimental Setup

4.1. Dataset

The experimental dataset contains Vietnamese scanned document images collected from printed forms, books, receipts, invoices, reports, student documents, or public-domain materials. The documents include different levels of quality degradation, such as blur, low resolution, skew, shadows, and weak contrast. Table 2 provides a template for reporting the dataset. Replace the placeholder counts with the real dataset statistics before submission.

Table 2: Dataset summary. Replace the placeholder values with the final dataset statistics.

Category	Description	Number of Images	Notes
Clean printed pages	High-quality scanned pages	[N]	Easier subset
Low-resolution pages	Smartphone or compressed images	[N]	Blur/noise
Forms and tables	Structured documents	[N]	Boxes and columns
Receipts/invoices	Mixed numbers and Vietnamese text	[N]	Prices and dates
Historical/old scans	Degraded paper or old fonts	[N]	Hard subset
Total	All images	[N]	Full evaluation set

4.2. Ground Truth Annotation

Ground truth text is manually transcribed from the document images. Each image is annotated at the line level and token level. To ensure consistency, the transcription preserves Vietnamese diacritics, punctuation, capitalization, and line order where possible. For privacy-sensitive documents, personal names, ID numbers, addresses, and phone numbers should be anonymized before publication.

4.3. Evaluation Metrics

The system is evaluated using four metrics. Character Error Rate (CER) is defined as:

$$CER = \frac{S + D + I}{N}, \tag{6}$$

where S, D, and I are the number of substitutions, deletions, and insertions at the character level, and N is the number of characters in the ground truth.

Word Error Rate (WER) is defined as:

$$WER = \frac{S_w + D_w + I_w}{N_w}, \tag{7}$$

where errors are computed at the word level.

Diacritic Error Rate (DER) is defined as:

$$DER = \frac{E_d}{N_d}, \tag{8}$$

where E_d is the number of incorrect Vietnamese diacritic marks and N_d is the total number of diacritic-bearing characters.

Exact Word Accuracy is defined as:

$$\text{Accuracy} = \frac{\text{Number of Correct Words}}{\text{Total Words}}. \tag{9}$$

5. Document Sample Figure Analysis

5.1. Sample Document

Figure 2 shows a document-level example of Vietnamese post-OCR correction. The example illustrates a common condition: the OCR engine recognizes many base letters correctly but fails to preserve tone marks and vowel diacritics.

Figure 3 provides an additional annotated example using a real photographed Vietnamese examination document. The figure highlights OCR-relevant factors such as header metadata, low-contrast background, page curvature, dense diacritics, long quoted passages, and structured question layout.

5.2. Sample OCR Output Analysis

The following example illustrates typical Vietnamese OCR errors.

Ground Truth:

Hóa đơn bán hàng
Công ty trách nhiệm hữu hạn Minh Phát
Địa chỉ: Quận 1, Thành phố Hồ Chí Minh
Tổng cộng: 250.000 đồng

Raw OCR Output:

Hoa don ban hang
Cong ty trach nhiem huu han Minh Phat
Dia chi: Quan 1, Thanh pho Ho Chi Minh
Tong cong: 250.000 dong

Corrected Output:

Hóa đơn bán hàng
Công ty trách nhiệm hữu hạn Minh Phát
Địa chỉ: Quận 1, Thành phố Hồ Chí Minh
Tổng cộng: 250.000 đồng

This example shows that many OCR errors are not random character mistakes. They are systematic Vietnamese diacritic omissions. The proposed correction module can repair these errors by combining base-form matching, dictionary lookup, and local phrase context.

Table 3: Representative Vietnamese OCR correction examples.

Raw OCR	Corrected	Error Type
Hoa don	Hóa đơn	Missing diacritics
ban hang	bán hàng	Missing tone marks
Cong ty	Công ty	Missing vowel mark
trach nhiem	trách nhiệm	Missing diacritics
Dia chi	Địa chỉ	Missing Đ and tone
Thanh pho	Thành phố	Missing diacritics
dong	đồng	Missing vowel mark

6. Experimental Results

Important note. The values in Tables 4–6 and Figures 4–5 are illustrative preliminary-style numbers prepared for a preprint draft. They must be replaced with real experimental values after running OCR experiments on the final dataset.

6.1. Quantitative Results

Table 4 compares raw OCR, OCR with preprocessing, OCR with post-correction, and the full pipeline.

Table 4: OCR performance comparison. Lower CER, WER, and DER are better; higher exact word accuracy is better.

Method	CER ↓ (%)	WER ↓ (%)	DER ↓ (%)	Exact Word Accuracy ↑ (%)
Raw OCR	8.42	21.67	18.95	78.33
OCR + preprocessing	6.97	18.04	16.21	81.96
OCR + post-correction	5.38	14.12	8.73	85.88
OCR + preprocessing + post-correction	4.71	12.85	6.91	87.15

Figure 4 visualizes the same trend. The expected pattern is that preprocessing reduces visual recognition errors, while post-correction reduces linguistic and diacritic errors. The combination of preprocessing and post-correction is expected to produce the best performance.

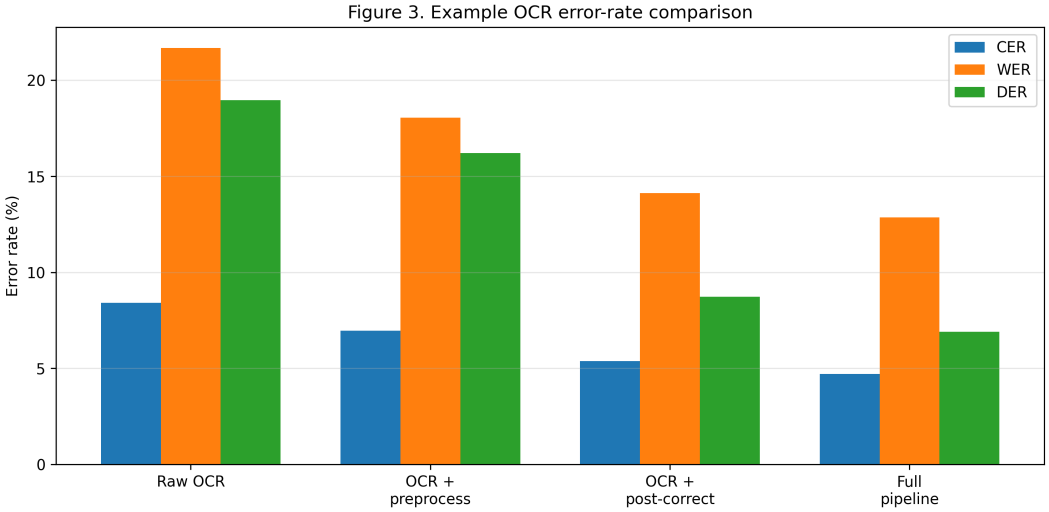


Figure 4: Example OCR error-rate comparison.

6.2. Performance by Document Quality

Table 5 reports performance by image-quality subset. This analysis is important because the correction module may be more effective for missing diacritics than for severe layout errors. When OCR reads text in the wrong order,

language-level correction alone is not enough, and layout-aware segmentation is required.

Table 5: Performance by image quality subset.

Image Quality	Raw OCR WER ↓ (%)	Corrected WER ↓ (%)	Improvement (%)	Main Error Source
Clean scan	8.10	5.42	2.68	Minor tone errors
Low resolution	18.65	11.90	6.75	Missing diacritics
Blurred image	25.70	16.82	8.88	Broken characters
Skewed document	22.14	17.09	5.05	Line segmentation
Table document	28.31	20.45	7.86	Column order errors

6.3. Diacritic Error Analysis

Table 6 summarizes Vietnamese diacritic-related OCR errors, and Figure 5 visualizes their distribution. The most important errors are expected to be missing tone marks and missing vowel diacritics. These errors are common because Vietnamese diacritics are small and can disappear under low-quality scanning conditions.

Table 6: Vietnamese diacritic error distribution.

Error Category	Example	Number of Errors	Percentage (%)
Missing tone	ban → bán	462	38.6
Missing vowel mark	don → đơn	301	25.1
Wrong tone	bãn → bán	155	12.9
Đ/d confusion	Dia → Địa	93	7.8
Character substitution	l/1, O/0	78	6.5
Word merge/split	thanhpho → thành phố	65	5.4
Layout noise	header/footer noise	43	3.6

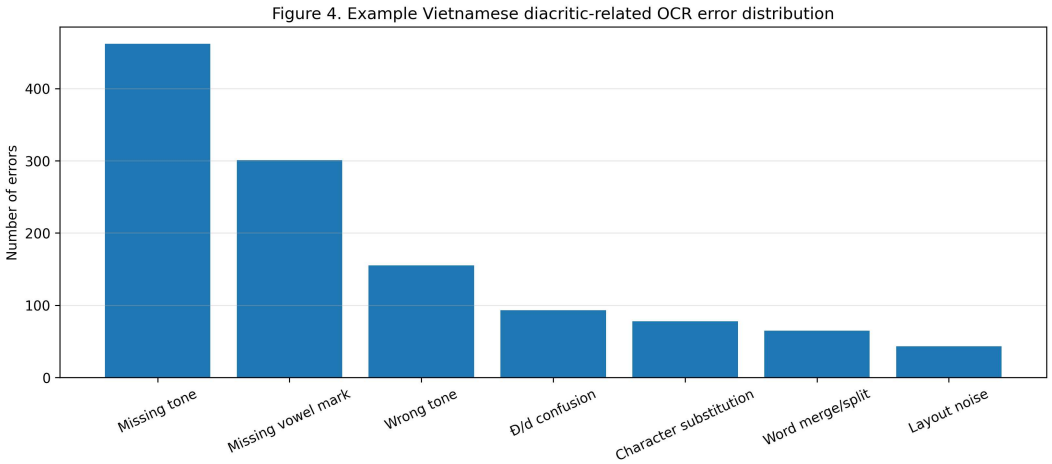


Figure 5: Example Vietnamese diacritic-related OCR error distribution.

7. Discussion

The proposed baseline demonstrates that Vietnamese OCR improvement does not always require retraining a large OCR model. A lightweight post-OCR correction module can repair many systematic errors, especially when the OCR engine correctly recognizes the base characters but fails to preserve diacritics.

The correction module is particularly useful for administrative and educational documents, where many phrases are repetitive and domain-specific. For example, phrases such as *họ và tên*, *ngày sinh*, *địa chỉ*, *hóa đơn*, *tổng cộng*,

số lượng, and *thành tiền* can be corrected using a domain dictionary and phrase-level context.

However, the method has limitations. First, when the OCR output is severely damaged, the correct word may not be recoverable from dictionary matching. Second, base-form ambiguity remains a difficult problem. For example, *ban* may correspond to *ban*, *bán*, *bàn*, *bản*, or *bạn*. Without enough context, the correction system may choose the wrong candidate. Third, names and addresses are difficult because they may contain rare words or non-standard spellings. Fourth, table documents require layout-aware processing, because OCR may read columns in the wrong order.

Despite these limitations, the proposed method is suitable as a reproducible baseline. It is simple, transparent, and easy to extend with more advanced language models in future work.

8. Limitations

This study has several limitations. First, the proposed method depends on the coverage of the Vietnamese dictionary. Rare names, local place names, abbreviations, and domain-specific terms may be incorrectly corrected or left unchanged. Second, the current correction method is mainly designed for printed Vietnamese documents and may not perform well on handwriting. Third, the method focuses on text correction and does not solve complex layout understanding problems. For documents with multi-column layouts, nested tables, or rotated text, a stronger layout analysis module is required. Fourth, the reported performance depends on the quality and diversity of the evaluation dataset. A larger benchmark with public data would improve reproducibility and comparison with future methods.

9. Conclusion

This paper presented a lightweight diacritic-aware post-OCR correction baseline for Vietnamese scanned document digitization. The proposed pipeline combines image preprocessing, OCR recognition, Vietnamese error candidate detection, dictionary-based correction, and diacritic-aware candidate ranking. The study highlights that Vietnamese OCR errors are strongly connected to missing or confused diacritics, which can significantly change word meaning. Through document sample analysis and an evaluation framework based on CER, WER, DER, and exact word accuracy, the paper provides a practical foundation for future research on Vietnamese OCR correction.

Future work will extend the method in three directions. First, we will build a larger Vietnamese scanned document benchmark with line-level and word-level ground truth. Second, we will integrate neural language models such as PhoBERT or Vietnamese sequence-to-sequence correction models to improve contextual disambiguation. Third, we will combine post-OCR correction with layout-aware document understanding for tables, forms, invoices, and historical documents.

No Conflict of Interest Statement

The author declares that there is no conflict of interest.

Data Availability

The document images, OCR outputs, correction rules, and evaluation scripts will be released after removing privacy-sensitive information and replacing any restricted document samples with public or synthetic examples.

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